# Fernanda Araujo Maciel, PhD

Management, Information Systems, and Business Analytics
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## EDUCATION

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|---|---|
| Ph.D. in Business, Business Analytics – Bentley University, Waltham, MA | **2020** |
| M.S. in Marketing Analytics – Bentley University, Waltham, MA | **2014** |
| M.S. in Business Administration and Marketing – PUC-Rio, Brazil | **2011** |
| B.S. in Statistics – Universidade Federal do Rio de Janeiro (UFRJ), Brazil | **2009** |

## PROFESSIONAL EXPERIENCE

**Assistant Professor** **2020–Present**
California State University, Sacramento, CA

- Teach undergraduate and graduate courses in analytics.
- Research in causal analysis, marketing analytics, and digital trust and technology.
- Supervise and mentor students and visiting scholars.

**Doctoral Fellow** **2016–2017**
Bentley University, MA

- Research Assistant working with Dr. Heikki Topi; managed six Master's students on "Exploring the Status of Education for Data Science."

**Market Research Analyst** **2014–2015**
Management Insight Technologies, Marlborough, MA

- Performed quantitative and qualitative market research for Amazon, AT&T, Cisco, Hewlett-Packard, and Microsoft.

**Analytics and Marketing Research Intern** **Jun–Dec 2013**
Copernicus Marketing Consulting and Research, Boston, MA

- Conducted Factor and Cluster Analyses for consumer segmentation projects for Red Sox, CVS, Panera Bread, and Hershey's.

**Additional Early Experience (Brazil)** **2006–2009**
Embratel, SulAmérica Seguros, and M4U

- Roles in marketing research, credit analysis, and business intelligence.

## RESEARCH AND PUBLICATIONS

Research Areas: Causal Analysis, Marketing and Retail Analytics, Digital Trust and Technology, Data-Driven Decision Making.

### Publications and Acceptances

Araujo Maciel, F. A., Xie, H. & Aivazpour, Z. "Patient Trust in AI-Powered Medical Assistants: Cognitive and Emotional Drivers Across Brazil and the United States". Computers in Human Behavior. [Conditionally Accepted, ABDC: A]

Araujo Maciel, F. (2026). "Unintended educational consequences of Bolsa Família program in Brazil". International Journal of Social Economics, Vol. 53 No. 9 pp. 1383–1398, https://doi.org/10.1108/IJSE-05-2025-0495 [ABDC: B, Acceptance Rate: 11.9%]

Carvalho, H., & Maciel, F. A. (2025). "RFM-Based Customer Segmentation: A Pedagogical Case Study for Marketing Analytics Education". Journal for Advancement of Marketing Education, 34(1), 14–26. https://doi.org/10.63963/001c.157562 [Acceptance Rate: 10.6%]

Maciel, F. A., & Duarte, D. (2023). "The impact of cash transfer participation on unhealthy consumption in Brazil". Health Policy OPEN, 3, 100087. https://doi.org/10.1016/j.hpopen.2022.100087

Maciel, F. A., da Rocha, A., & da Silva, J. F. (2013). “Brand personality of global quick-service restaurants in emerging and developed markets: A comparative study in Brazil and the US”. Latin American Business Review, 14(2), 139–161. DOI: 10.1080/10978526.2013.808950

**Under Review**

Araujo Maciel, F. A. & Carvalho, H. Customer Segmentation and Personalized Promotions in Fuel Retailing: A Temporal and Fuzzy Profiling Approach. Minor Revision at the Journal of Business Research.

Shenoy, P., Stark, R., Maciel, F., Owens, M., & Blake, S. Reported Barriers to Librarian and Nursing Faculty Collaboration in the Development and/or Implementation of OER Materials in Nursing Courses. Under Review at Evidence Based Library and Information Practice.

**In Preparation**

Aivazpour, Z. & Maciel, F. A. Drowning in Information: A Mixed-Methods Study on the Effects of Distrust and Fake News During Elections. Expected submission Fall 2026.

Aivazpour, Z. & Maciel, F. A. From Crisis to Post-Crisis: The Evolution of Vaccine-Skeptical Discourse on Social Media (Evidence from X and Reddit, 2021–2026). Expected submission Fall 2026.

Carvalho, H., Maciel, F. A., & Severo, M. Conditional and Increasing Long-term Effects of Loyalty on Price and Marketing Performance. Expected submission Fall 2026.

Maciel, F. A. Eating patterns among food-secure and food-insecure households in Brazil: An exploratory analysis of food co-consumption.

Aivazpour, Z. & Maciel, F. A. AI Adoption in Healthcare: Challenges Under the Proposed HIPAA Security Rule.

Carvalho, H. & Maciel, F. A. Mandatory Price Disclosure, Private Loyalty Discounts, and Competitive Pricing Errors in Fuel Retail: Evidence from Brazil.

Carvalho, H. & Maciel, F. A. The Instrument Is the Model: AI-Generated Constructs and the Silent Return of the Imposed Etic in International Marketing. Expected submission Spring 2027.

Blake, S., Owens, M., Maciel, F., Shenoy, P., & Stark, R. Implementing OER Materials in Nursing Curricula: Understanding Barriers to Faculty-Librarian Collaboration. *Nurse Education*

Aivazpour, Z. & Maciel, F. A. Does Generative AI Undermine Students’ Independent Learning? Insights from Vygotsky’s Constructivist Theory.

**Non-Academic Publications**

Maciel, F. A. (2024). Do I really need to do Descriptive Statistics? In H. Anunciação (Ed.), The Data Scientist and his Demons (pp. 38–44). CX Trilogy. ISBN-13: 979-8302203915.

## SELECTED RESEARCH PRESENTATIONS

Aivazpour, Z. & Maciel, F. A. (2026). Does Generative AI Undermine Students’ Independent Learning? Insights from Vygotsky’s Constructivist Theory. Accepted at ICIS 2026 (*To be presented in December*).

Maciel, F.A (2026). From Data to Decisions: Teaching Customer Analytics with Real Data. Innovation Lab session at the DSI Annual Pedagogy Conference, online, May 2026. [Invited]

Shenoy, P., Stark, R., Maciel, F., Owens, M., & Blake, S. (2026). Cultivating collaboration: Understanding the barriers between librarians and nursing faculty in the development and use of OER. MLA Annual Conference, Milwaukee, WI, May 2026. (Presented by P. Shenoy and R. Stark.)

Maciel, F. A. & Carvalho, H. (2026). Extending RFM with Temporal and Volume Dimensions for Customer Segmentation and Promotion in Fuel Retailing. WDSI Annual Meeting, Vancouver, BC, March 2026.

Severo, M., Carvalho, H., & Maciel, F. A. (2025). Conditional and permanent effects of loyalty engagement on sales volume, price, and transaction size. EnANPAD, Aracajú, Brazil, October 2025. (Presented by M. Severo.)

Maciel, F. A. & Xie, H. (2025). AI in Healthcare: Patients’ Perspectives on AI-Powered Medical Assistant. WDSI Annual Meeting, Honolulu, HI, April 2025.

Maciel, F. A. (2024). Cash Transfers and Nutritional Wellbeing: A Brazilian Case Study Using Machine Learning and a Social Determinants Framework. Industry Studies Association Annual Conference, Sacramento, CA, June 2024.

Maciel, F. A. (2024). Evaluating Food Consumption Patterns in Brazil's Low-Income Population. WDSI Annual Meeting, Sonoma, CA, April 2024.

Maciel, F. A. (2022). Unintended Consequences of Welfare Programs in Schooling Outcomes: Evidence from Brazil. WDSI Annual Meeting, Waikoloa, HI, April 2022.

Maciel, F. A. (2021). Does the Brazilian Bolsa Familia Program reduce Child Labor? INFORMS Annual Meeting, Anaheim, CA, October 2021.

## TEACHING EXPERIENCE

**California State University, Sacramento**

Undergraduate

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|---|---|
| DS 101 – Introduction to Business Analytics | **F20–Present** |
| DS 102 – Foundation and Tools for Business Analytics | **S24–Present** |
| BHON 106 – Business Data Analysis [Business Honors Program] | **F24** |
| MIS 183 – Business Intelligence Applications | **F23** |

Graduate

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|---|---|
| BUS 802 – Statistics Boot Camp [online & in-person] | **S21, F21, S22, F25, F26** |
| MSBA 205 – Data Analytics for Business [hybrid] | **Su23, S24, Su25** |
| iMBA 214 – Statistical Decision Making [online] | **S22, S25** |
| EMBA 223 – Quantitative Methods for Decision Making | **S23, F23** |

**Bentley University – Undergraduate**

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| MA 214 – Intermediate Applied Statistics | **F19–S20** |
| GB 213 – Business Statistics | **F17–S19** |

**Curso de Capacitação em Estatística (online)** **2021–Present**

Instructor and creator of a Portuguese-language online statistics course for Brazilian learners with over 1700 students. Provide ongoing student support and facilitate live sessions featuring student projects with real-world datasets.

## EDUCATIONAL INNOVATION

**Course Design**

**MCM 202 – Applied Statistics for Cybersecurity Management** **Spring 2026**

Designed and developed a graduate-level applied statistics course for the new Master's in Cybersecurity Management (MCM) program at Sac State. The program is currently in the approval process for launch.

**DS 102 – Foundation and Tools for Business Analytics** **Spring 2024**

Designed and implemented a hands-on course focused on data preparation and analysis using R and Python.

**Curso de Capacitação em Estatística 2.0 – Brazil** **2024**

Redesigned, re-recorded, and expanded the original 40-hour course into an 80-hour Portuguese-language statistics course, currently offered as a university extension course in partnership with a Brazilian higher-education institution and certified by the Brazilian Ministry of Education (MEC).

**AI-Enhanced Pedagogy** **2024-Present**

Developed experiential, AI-enhanced assignments in which students learn to use AI prompts to generate Python code and create data visualizations. Emphasizes iterative prompting, critical thinking, and reflection on AI-assisted coding workflows.

## PUBLIC ENGAGEMENT AND MEDIA

**Digital Outreach**

**YouTube Channel: Prof. Fernanda Maciel** **2020–Present**

Creator of a Portuguese-language YouTube channel focused on statistics and analytics, with over 980,000 views, 72,000 hours of watch time, and 23,600 subscribers.

**Instagram: Prof. Fernanda Maciel** **2020–Present**

Professional educational content creator with over 65,000 followers, developing Portuguese-language content on statistics, data analytics, and data-related careers.

**LinkedIn: Fernanda Maciel, PhD** **2021–Present**

Professional content creator with over 28,000 followers, sharing educational content on statistics, analytics, higher education, and professional development

**Estatisticast Podcast** **2023–2024**

Founder and host of a podcast available on YouTube and Spotify covering real-world data applications and analytics careers. Reached over 53,000 YouTube views and 68,000 Spotify streams, totaling over 24,000 consumption hours and 5,600 followers.

**Press Coverage**

**American Statistical Association (ASA).** *How I Built a Portuguese-Language Analytics Community Online. ASA Magazine*, August 3, 2026. Feature article on my YouTube channel, the Estatisticast podcast, and my initiatives to expand access to statistics and data analytics education in Portuguese. https://magazine.amstat.org/blog/2026/08/03/portuguese-language-analytics-online/

**San Francisco Business Times.** *Sacramento State's Fernanda Maciel, Ph.D., expands access to applied analytics education.* March 1, 2026. Feature article on my educational initiatives in business analytics and online learning. https://www.bizjournals.com/sanfrancisco/news/2026/03/01/sacramento-state-maciel-access-applied-analytics.html

**Invited Talks – Industry, Community, and Online Platforms**

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| Mentoring – Mulheres e Tecnologia (Women and Tech) | **2025** |
| Laboratoria NGO for Women in Tech | **2024** |
| SQL Week Recife (Conference) | **2024** |
| Nubank Data Science & Machine Learning Meetup | **2023** |
| BIX Tecnologia – Semana de Dados | **2022, 2023** |
| Data Science Week, ABRACD | **2022** |
| Data Girls Neuron – (re)Start Data Science | **2021** |
| 5th Brazilian CONE (Congresso de Estatística) | **2021** |

## ACADEMIC INVITED TALKS

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| Seminar Session – PUC-Minas, Brazil | **2026** |
| Academic Week CCSA – Universidade Católica de Petrópolis, Brazil | **2023** |
| Seminar Session – Candido Mendes University, Brazil | **2023** |
| Opening Lecture, MBA in Data Science – University of São Paulo, Brazil | **2021** |

## SUPERVISION AND MENTORSHIP

**Faculty Advisor – Future Business Leaders of America–Phi Beta Lambda (FBLA-PBL)** **2025–Present**

**Faculty Mentor**

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| Research Poster Presentation – Rahul R. Wamanacharya (MSBA) | **2025** |
| Data Mine Network Program (NSF-funded) – Laura M. Young | **2024** |

**Visiting Scholar Supervision**

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| Dr. Henrique Carvalho – Consumer behavior and loyalty programs in retail analytics | **2024–2025** |

## SERVICE

**Department Service**

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| Department Chair Nomination Committee | **2026** |
| Undergraduate Advisor | **2025–Present** |
| Faculty Qualifications Review Committee, chair | **2024–Present** |

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| Decision Sciences/Business Analytics Search Committee, chair | **2024–2025** |
| Part-Time Appointment Committee, chair | **2023–2025** |
| Applied Statistics course development for the MSM Program | **2024–2025** |
| ISBA Fellows Meeting | **2023–2025** |
| New course proposal – DS 102: Foundation and Tools for Business Analytics | **2022** |
| Informs Career Fair Informational Interviewer | **2021** |
| Decision Sciences/Business Analytics Search Committee | **2021** |

**College Service**

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| Academic Programs Committee (Faculty Council) | **2026–2028** |
| Faculty Development & Awards Committee | **2026–2028** |
| External Programs Committee (Faculty Council) | **2024–2028** |
| Peer Reviewer, Research Enhanced Support Grant Program | **2024–2025** |
| Grove Consultation Workshops Participation | **2024** |
| MBA Programs Application Interviewer | **2023–2024** |
| Assessment Committee | **2021–2024** |
| COB Scholarship Committee | **2020–2024** |
| AACSB Review Visit – Untenured Faculty Meeting | **2022** |

**University Service**

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| Invited Facilitator, CTL Summer Teaching Institute | **2026** |
| Grade Appeal Panel Committee | **2026–2028** |
| Exceptional Assigned Time Subcommittee | **2026–2028** |
| Pedagogy Enhancement Award (PEA) Subcommittee, panel chair | **2024–2027** |
| Research and Creative Faculty Awards Reviewer | **2025, 2026** |
| Faculty Advisor, Future Business Leaders of America–Phi Beta Lambda | **2025** |
| Student Research and Creative Activity Symposium Juror | **2024–2025** |
| Research and Creative Faculty Awards Reviewer | **2022–2023** |
| Wang Family Excellence Award Committee | **2021–2022** |

**Community Engagement**

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| Councilor, ASA Sacramento Statistical Association (SSA) | **2026–Present** |
| Statistical expert for cross-disciplinary research partnership | **2025–Present** |
| Workshop presenter, Data\|Tech Lab at Sac State | **2025, 2026** |
| Volunteer data consulting, Student-Run Physical Therapy Clinic | **2024–2026** |
| Faculty Juror, Student Research and Creative Activity Symposium | **2024–2026** |
| Judge, ASA DataFest Students Challenge Competition at Sac State | **2023, 2025** |
| Analytics presentation to Twin River Middle School students | **2023, 2024** |
| Hornet Bus Tour participation | **2024** |
| Motherscholar Zine Project collaboration | **2024** |
| Subject matter expert, Sacramento Hispanic Chamber of Commerce | **2021** |

**Conference Reviewer**

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| Western Decision Science Institute (WDSI) | **2024–Present** |
| International Conference on Information Systems (ICIS) | **2024** |
| Hawaii International Conference on System Sciences (HICSS) | **2018–2022** |

**Professional Memberships**

American Statistical Association (ASA)
Association for Information Systems (AIS)
Decision Sciences Institute (DSI)
Institute for Operations Research and the Management Sciences (INFORMS)
National Center for Faculty Development and Diversity (NCFDD)

## PROFESSIONAL DEVELOPMENT

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| North Far North Data Analytics and Digital Innovation Regional Convening | **Oct 2026** |
| Summer Research Academy | **Jun 2026** |
| CTL ACUE Internships: Inspiring Inquiry and Preparing Lifelong Learners | **Summer & Fall 2026** |
| DSI Annual Pedagogy Conference (online) | **May 2024, 2026** |
| ORIED Research Data and Publishing | **2025** |
| CTL Lunch and Learn Series | **2025** |
| CTL ACUE Generative AI: Teaching and Learning with AI | **2024** |
| Causal inference is not a statistical problem – Lucy McGowan (UCL/UNIL) | **2023** |
| Causal Inference with Cross-Sectional Data – Jeffrey Wooldridge (instats) | **2023** |
| CTL Summer Teaching Institute | **2022** |
| Certificate Program in Student Success Analytics | **2022** |
| Industry Studies Association (ISA) Professional Development Workshop | **2021** |
| ACUE Course Promoting Active Learning Online | **2020–2021** |
| Wellness in the Classroom | **2020–2021** |
| Demystifying the Tenure-Track | **2020–2021** |
| NISS Virtual Career Fair for Statisticians in High-Tech Industries | **2021** |
| Summer Camp Teach ON!-Line | **2020** |

## AWARDS AND HONORS

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| Outstanding Faculty Award for Community Service, College of Business | **2025–2026** |
| ISA Professional Development Workshop | **2021** |
| PhD Fellowship, Bentley University | **2015–2020** |
| PhD Consortium Participant, DSI Annual Meeting | **2019** |
| Research Assistant, National Science Foundation (NSF) | **2017–2018** |
| McCallum Leadership Scholar Award, Bentley University | **2012–2014** |
| Research Grant, National Council for Scientific and Tech Development (CNPq) | **2009–2011** |